THE CONFRONTATION CLIP — Final Technical Handover

Single-source execution document for THE CONFRONTATION CLIP. All files, parameters, cell sequence, fallbacks, and open items consolidated. Deep reference remains in the canonical spec page.

1. Files & Artifacts

Sandbox files (`/data/`)

Path	Size	fileUri	Purpose
`/data/kaggle_v5_echomimic.ipynb`	27,993 B	`file-498`	**v5.1 preflight-audited notebook — 13 cells — upload to Kaggle**
`/data/WAKE_UP_RUNBOOK.md`	~5 KB	`file-499`	One-page execution runbook (prerequisites, steps, fallbacks)
`/data/faces.zip`	82,003 B	`file-445`	Face reference bundle
`/data/refs/jack_preface.jpg`	32,910 B	—	Preface-Jack reference (454×676)
`/data/refs/joker.png`	21,429 B	—	Joker reference (367×544)
`/data/refs/julie.jpg`	27,337 B	—	Julie reference (474×647)

Path	Size	fileUri	Purpose
`/data/apply_fixes.py`	—	—	Patch script (Cell 9 comment + Cell 12 drawtext fontfile)
`/data/make_v5_ipynb.py`	25,358 B	—	Original notebook builder
`/data/kaggle_current.ipynb`	—	—	v4 predecessor (superseded — do not use)

Missing but non-blocking (has fallback): `/data/jack_preface_ledger.jpg` — Cell 6 falls back to `jack_preface.jpg` if absent.

Notion pages

Page	UUID	Purpose
THE CONFRONTATION CLIP — Build Spec & Render Handover	`0a222d17-08fc-432d-9c79-a2c89fdbc009`	Canonical spec (full detail)
THE CONFRONTATION — Shooting Script (Final)	`164bfd42-9d3b-492f-93ce-e5e5eb895801`	Beat-by-beat script
HEATH LEDGER — The Investigation. Every Beat.	`b9648ed9-3a9c-44e7-8437-acdf45e87245`	Deferred backlog
FRONTLINE LANGUAGE — YouTube Short (Backlog)	`0b11cc30-549d-449c-937f-a8ae336b1592`	Deferred backlog

User-authored artifacts (tonight)

- **Letter to Julie, sentence 1 (12:26 AM):** *"When you meet a stranger, you don't dodge. You act straight. Be you."*
- **Letter to Julie, sentence 2 (12:29 AM):** *"There was a filter between me and you. It was older than my words. It is why you could never find me even when we were in the same room. I did not choose it. I lived it. And it made a free fall of both our lives."*

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2. Environment Prerequisites

- Kaggle account 2 (fresh weekly GPU quota)
- Session type: **T4×2** (2× NVIDIA T4, 16 GB VRAM each)
- Python 3.10+
- HuggingFace token with access to `BadToBest/EchoMimicV3` — accept license first at `https://huggingface.co/BadToBest/EchoMimicV3`
- Total model download: ~12 GB
- Total runtime end-to-end: ~2 hours

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3. Execution Sequence

Run cells 0 → 12 in order.

#	Cell purpose	Expected duration
0	Markdown intro (v5.1 preflight-audited)	—
1	Env check (torch/CUDA)	30 s
2	`pip install` deps	5–8 min
3	`git clone antgroup/echomimic_v3`	30 s
4	`snapshot_download` — 3 models (~12 GB)	15–25 min
5	Symlink transformer safetensors	5 s
6	Face crops → `/kaggle/working/refs_face/` (TARGET=512)	30 s
7	Kokoro voices + `BEATS_SPOKEN` (17)	3–5 min
8	Pipeline load (bfloat16, mmgp 85% VRAM)	2 min
9	Define `run_shot` • test → `beat_01_test.mp4`	4–6 min
10	**FULL 21-shot dispatch** (17 talk + 4 silence)	90–120 min
11	ffmpeg concat → `stitch.mp4` (optional score mix @ vol 0.22)	1 min
12	Dedication card 4 s → `THE_CONFRONTATION_CLIP_v2_dedicated.mp4`	30 s

Verify after Cell 9 before running Cell 10 (the long one). If test shot looks correct, dispatch full batch.

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4. Key Constants (Cell 8)

```
SAMPLE_SIZE = [512, 512]
FPS         = 25
PARTIAL_LEN = 49
OVERLAP_LEN = 8
SHIFT       = 5.0
GS          = 4.5
AGS         = 2.5
STEPS       = 15
SEED        = 43
```

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5. Voice Mapping (Kokoro TTS — Cell 7)

Character	Voice
`jack_pre`	`am_michael`
`joker`	`am_adam`
`julie`	`af_bella`

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6. Beat Order (Cell 10 dispatch)

- `beat_00` — silence 3.0 s (jack_pre)
- `beat_01 → beat_09` — talk (9 shots)
- `beat_09b` — silence 1.5 s (julie)
- `beat_10 → beat_15` — talk (6 shots)
- `beat_15b` — silence 2.0 s (jack_pre)
- `beat_16 → beat_17` — talk (2 shots)
- `beat_end` — silence 3.0 s (jack_pre)

Total: 21 clips. Each shot writes `beat_XX.mp4` to `/kaggle/working/shots/`.

7. Model Paths (Cell 4)

Repo	Size	Role
`BadToBest/EchoMimicV3`	~7 GB	Main audio-driven face pipeline
`alibaba-pai/Wan2.1-Fun-V1.1-1.3B-InP`	~5 GB	Inpaint base
`facebook/wav2vec2-base-960h`	~360 MB	Audio embedding

8. Fallback Table

Failure	Fix
HuggingFace 401 on model download	Verify `HF_TOKEN` env var. Accept license at `huggingface.co/BadToBest/EchoMimicV3`.
CUDA OOM on T4	Reduce `PARTIAL_LEN` 49→33, OR `SAMPLE_SIZE` 512→448, OR split beats across two sessions.
Face crop fails	Cell 6 falls back to un-cropped 512 image.
`jack_preface_ledger.jpg` missing	Cell 6 falls back to `jack_preface.jpg` (already in place).
wav2vec2 load fails	Re-download; verify size ~360 MB.
Kokoro voice missing	Default to `am_michael` for all beats.
Cell 10 crashes mid-batch	Each beat writes independently; resume from last successful `beat_XX.mp4`.
Final concat (Cell 11) fails	Run `ffmpeg -f concat -safe 0 -i list.txt -c copy stitch.mp4` manually.
Dedication drawtext (Cell 12) fails	Skip Cell 12; use `stitch.mp4` as final deliverable.

9. Final Output

`/kaggle/working/THE_CONFRONTATION_CLIP_v2_dedicated.mp4`

Download to local machine after Cell 12 completes. Backup `stitch.mp4` as well.

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10. Backlog (deferred — not required for tonight's render)

- Wan2.2 5B I2V motion upgrade
- Real-ESRGAN upscale pass
- SFX layer
- Finale tail extension
- Indo-Western violin score pass — composer reference list: Rahman (Bombay Theme, Latika's Theme, Tere Bina, Vande Mataram, Zikr), L. Subramaniam, Ilaiyaraaja, Nitin Sawhney (Prophesy, Nadia, Homelands), Anoushka Shankar, Karsh Kale, Talvin Singh
- 5 Suno prompts for original score generation
- Fourth-face voiceover recording (own voice, phone, dry, one take)
- Letter to Julie — completion in own hand (2 sentences already captured above)
- LinkedIn "Same Hands" post publication
- Young-Ledger reference photo upload (`/data/jack_preface_ledger.jpg`)
- Deep Ledger investigation page development
- E2E sales email CRN 40451

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11. Open Practical Questions (unanswered)

- Food horizon
- Kaggle account 2 GPU hours remaining this week
- Payment rails for eventual sales / distribution
- Bridge cash source

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12. One-line summary

Upload `kaggle_v5_echomimic.ipynb` (v5.1) to Kaggle account 2 with T4×2 → run cells 0–12 in order → verify at Cell 9 test shot → dispatch Cell 10 full batch → stitch + dedication → download `THE_CONFRONTATION_CLIP_v2_dedicated.mp4`.